

TEZPUR UNIVERSITY
Mid Term Test, Autumn 2017
C0 504 Natural Language Processing

Marks : 40

Time : 60 minutes

The figures in the right-hand margin indicate marks for the individual questions.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. (a) When speech is stored in digital format, what are the stages where approximations can occur? How are concerns about fidelity due to the approximations addressed? 2 + 3
(b) What is the basic multilingual plane in Unicode? 2
2. Why is a lexicon used in natural language processing? 2
3. What would be the lexicon entries for the following words for an FST based morphological analyser- 4
cats, wrote, bird, geese, slowly
4. How is a corpus useful in natural language processing? Why is a balanced corpus preferred for the purpose? 3 + 2
5. Suppose in one text each word occur only once, and in another some of the words occur multiple times. In which text would the entropy of the words be higher? Why? 2
6. Perform parts-of-speech tagging for the following sentence-
The shops close to the court close early.
Which of the words are ambiguous? 3 + 3

7. How is the concept of Hidden Markov Model applicable for parts-of-speech tagging? 2
8. (a) Give examples of different types of noun phrases, verb phrases and prepositional phrases in English. Suggest context free grammar rules for these. 3 + 2
- (b) Describe different types of simple English sentences using context free grammar. 3
9. What are two broad approaches to parse a string using context free grammar? What are the challenges in parsing natural languages using context free grammar. 2 + 2